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ENGINEERING SERVICES · INTERNAL ANALYSIS

NEXUS HUBBARD · HUBBARD, TX

Engineering Budget Scenario Analysis.

A side-by-side comparison of $500K and $1M engineering services engagements under the MaiaEdge/Pearce LNTP scope -- with projected 90-day outcomes, procurement readiness, and GMP conversion risk for the Nexus Hubbard multi-building connectivity architecture.

Prepared by MaiaEdge Networks · Pearce Services Group · May 2026

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01 ENGAGEMENT OVERVIEW

Framing

The LNTP scope as written commits to six distinct deliverable categories across a 12-week engagement: baseline architecture assumptions, Phase 1 LLD for four buildings, a procurement-ready BOM with long-lead identification, an expansion architecture to 65 buildings, interface coordination across carriers and contractors, and a construction-ready handoff package that supports GMP conversion.

The budget cap determines how many qualified engineers we can field, for how long, and how deeply we can work each deliverable. The scenarios below are built on the rate assumptions and staffing models described in the methodology note.

Methodology Note

The analysis uses a blended bill rate of approximately $185/hr for the combined MaiaEdge/Pearce engineering team, reflecting a mix of principal architects, senior network engineers, design engineers, and project management. At that rate, $500K purchases roughly 2,700 billable hours and $1M purchases roughly 5,400 hours. Travel and out-of-pocket expenses are excluded from hour counts and assumed to consume 5-8% of each budget.

02 BUDGET SCENARIO A

Scenario A: $500,000 Cap

Team Composition (12 weeks)

At $500K, after reserving approximately $35K for travel and third-party costs, the billable hour pool is roughly 2,500 hours:

- 1 Lead Network Architect (50% time)
- 1 Senior Design Engineer (75% time)
- 1 Project Manager / Coordination Lead (50% time)
- Pearce ISP/OSP contribution (limited to 20% time, ~1 engineer)

This is a lean two-person-equivalent engagement. It can move fast on a narrow scope but has very limited capacity for parallel workstreams.

Deliverables by Day 90

Baseline Architecture Document (T+2)

Achievable at $500K. Kickoff, technical input request, and baseline assumptions document are not budget-sensitive. This milestone is not at risk.

Phase 1 LLD -- Four Buildings (T+4)

Achievable, but with constraints. MMR and carrier hotel architecture can be designed to a competent LLD. Per-building connectivity and interbuilding fiber mesh will be completed, but depth of documentation and peer review is limited. No budget cushion for significant rework. If Owner inputs arrive late, this milestone slips and the remaining schedule compresses.

BOM, MTO, and Long-Lead Identification (T+6)

This is where $500K begins to show strain. A well-structured BOM covering fiber distribution panels, ODF frames, rack specifications, PBC hardware, and cabling can take several hundred hours at the component level. The BOM at T+6 will be a high-confidence estimate of major categories and quantities -- not a line-item procurement-ready package. Long-lead items can be flagged and ranked by lead time.

Expansion Architecture to 65 Buildings (T+8)

At $500K, this becomes a reference architecture document with conceptual fiber expansion models rather than a fully engineered buildout plan. The scalable topology and modular template will be defined, but capacity planning assumptions will be directional rather than validated by detailed design work.

Construction-Ready Handoff Package (T+12)

The Phase 1 handoff package will be structurally complete, but the cost basis supporting GMP conversion will carry wider uncertainty bands. GMP conversion at 60% design is possible, but depends heavily on the quality and timeliness of Owner-provided design inputs. A two to three week input delay puts the T+12 package at the GMP table with gaps.

Day 90 State at $500K

Phase 1 four-building LLD is complete and defensible. Long-lead item categories are identified but not fully spec'd for purchase orders. The expansion architecture is conceptual. GMP conversion is possible but the cost basis carries risk. Procurement for long-lead items can begin, but PO placement will likely trail the 90-day mark by four to six weeks.

> Risk: If Nexus confirms nine buildings as the working basis (Project Apex), $500K is insufficient. The engagement either narrows to a subset of the nine-building scope or requires a budget amendment before reaching T+6.

03 BUDGET SCENARIO B

Scenario B: $1,000,000 Cap

Team Composition (12 weeks)

At $1M, after reserving approximately $60K for travel and third-party costs, the billable hour pool is approximately 5,100 hours:

- 1 Lead Network Architect (full time)
- 2 Senior Design Engineers (full time)
- 1 Pearce ISP/Physical Layer Lead (full time)
- 1 Project Manager / Coordination Lead (75% time)
- 1 Technical Documentation Engineer (50% time)

This is effectively a five-person engagement with parallel workstreams running simultaneously. Design, documentation, and coordination proceed in parallel rather than sequentially.

Deliverables by Day 90

Baseline Architecture Document (T+2)

Same as Scenario A. Not budget-sensitive.

Phase 1 LLD -- Four Buildings (T+4)

Delivered with greater depth. The LLD package includes the full MMR and carrier hotel architecture, per-building connectivity design, interbuilding fiber mesh, and a detailed interface block diagram showing responsibility handoffs by contractor. Design review cycles with Nexus, Michels, and AppWell can be accommodated without compressing the schedule. If the Anthropic building plans are used as a design template, the LLD for buildings two through four can be parallelized with the lead architect's review rather than serialized.

BOM, MTO, and Long-Lead Identification (T+6)

This is the most meaningful difference between the two scenarios. At $1M, the BOM at T+6 is a genuine procurement-ready package: component-level for major categories, with manufacturer and model specifications, estimated quantities per building, and lead time ranges by item. The Pearce physical layer lead builds the ISP BOM in parallel with the MaiaEdge team building the network active equipment BOM. Long-lead items are not just flagged -- they come with recommended order timing relative to the Q4 2026 / Q1 2027 MMR go-live target. This is the output that allows Nexus to place purchase orders inside the 90-day window.

Expansion Architecture to 65 Buildings (T+8)

The expansion document at $1M is an engineered reference architecture, not a narrative framework. It includes a validated fiber expansion model, buildout templates at the per-building level, a capacity planning model with assumptions and sensitivities, and modular design patterns Nexus can hand to any future engineering team. This document becomes a durable asset for investor and lender diligence.

Construction-Ready Handoff Package (T+12)

The Phase 1 handoff package is complete and GMP-ready. The cost model carries defensible unit costs validated against current vendor pricing from the BOM work. Per-building installation duration estimates are grounded in the physical design rather than rule-of-thumb benchmarks. GMP conversion at 60% design -- Nexus's stated threshold -- is achievable within the 90-day window.

Day 90 State at $1M

Phase 1 four-building LLD is complete and construction-ready. The BOM for long-lead items is procurement-ready, and Nexus can place purchase orders with reasonable confidence during Q4 2026. The expansion architecture is engineered to a level that supports investor and lender diligence. GMP conversion is on track. If Nexus confirms nine buildings as the working basis, the $1M envelope can absorb a modest scope expansion on the expansion architecture deliverable without amending the agreement, provided the Phase 1 LLD scope remains at four buildings.

> Risk: The $1M scenario still depends on receiving Owner design inputs at or near engagement kickoff. A three-week delay in the Anthropic building plans or MMR specifications will compress the T+4 through T+6 deliverables. At five engineers, there is more capacity to absorb that compression -- but it is not unlimited.

04 COMPARISON MATRIX

Side-by-Side Comparison

Deliverable / Outcome	$500K Result	$1M Result
Baseline Architecture (T+2)	Complete	Complete
Phase 1 LLD -- 4 buildings (T+4)	Complete; limited review cycles	Complete; parallel workstreams, peer-reviewed
BOM / Long-Lead ID (T+6)	Category-level estimates only; not procurement-ready	Component-level, procurement-ready; POs can be placed
Expansion Architecture (T+8)	Conceptual framework; directional assumptions	Engineered reference architecture; investor-grade
Construction-Ready Handoff (T+12)	Structurally complete; wider cost uncertainty	GMP-ready; defensible unit costs
GMP Conversion at 60% Design	Possible; cost basis carries risk	On track; cost basis is defensible
9-Building Scope (Project Apex)	Not feasible without budget amendment	Manageable on expansion deliverable
Long-Lead PO Timing	4-6 weeks after Day 90	Within Day 90 window

05 RECOMMENDATION

Recommendation

The $1M engagement is the right structure for this project. The practical difference between the two scenarios is not effort level -- it is whether Nexus can place long-lead purchase orders during the 90-day window or six weeks after it. That timing gap directly affects their Q4 2026 / Q1 2027 MMR go-live target.

Given that Nexus has stated the engagement is schedule-driven rather than budget-driven, the additional $500K is best understood as schedule insurance, not cost padding.

When $500K is defensible:

- Nexus formally confirms a strict four-building scope (no Apex expansion)
- Owner design inputs -- Anthropic building plans, MMR dimensions, confirmed power/cooling specs -- are delivered at or before Day 1
- Nexus accepts a BOM that requires one to two additional weeks of refinement before supporting procurement-ready purchase orders

When $500K is not defensible:

- The working basis shifts to nine buildings (Project Apex)
- Owner design inputs arrive with any meaningful delay
- Nexus expects a procurement-ready BOM within the 90-day window

In either scenario, the budget should be structured as a not-to-exceed with monthly cost reporting and an 80% drawdown notification, as the LNTP draft prescribes. The actual spend may come in below the cap -- particularly if the Anthropic cookie-cutter design simplifies per-building LLD work -- but holding the cap at $1M preserves optionality if scope expands or input delays create rework cycles.